Evaluation of a high-recall search engine, Variomes, to curate oligogenic diseases variants

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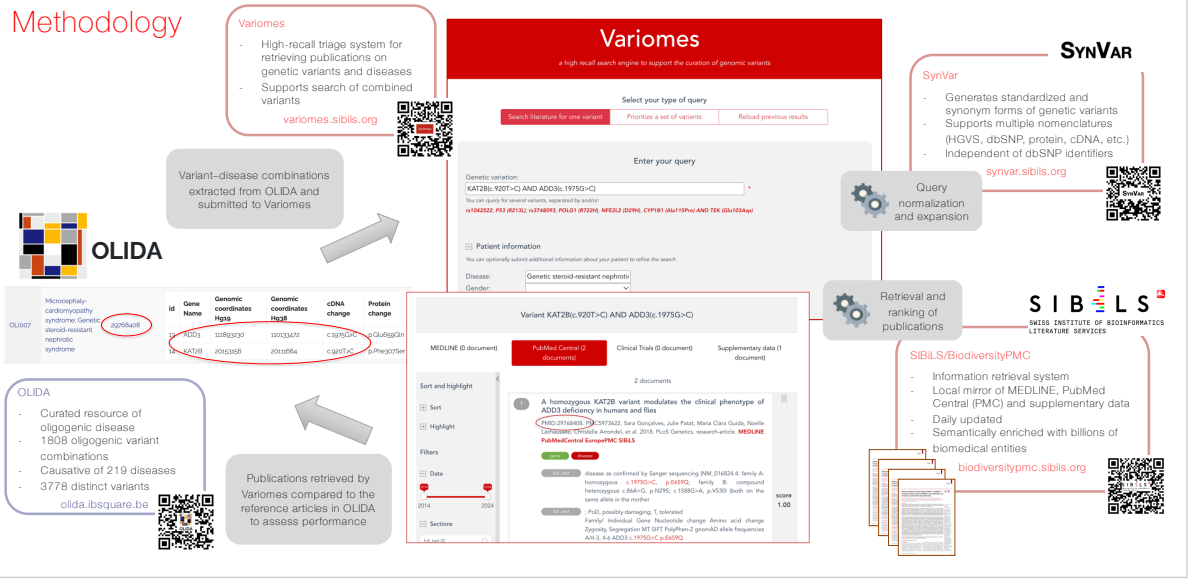
Introduction

- Oligogenic diseases are caused by the combined effect of multiple genetic variants
- Their interpretation is challenging due to limited and scattered literature
- OLIDA is a manually curated database of oligogenic diseases, based on scientific publications
- High-recall tools are essential to support the curation of oligogenic diseases

Methodology

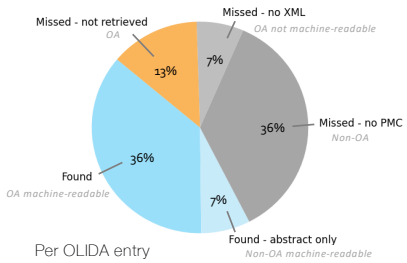


Objectives

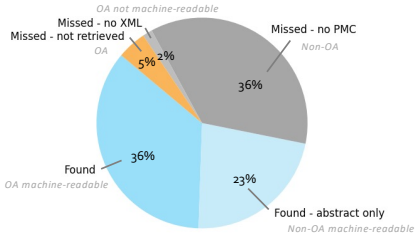
- Evaluate Variomes, a high-recall literature triage system for retrieving publications on genetic variants and diseases, using OLIDA as a reference benchmark
- Identify current limitations and support improvements in the automatic retrieval of literature on oligogenic diseases

Results

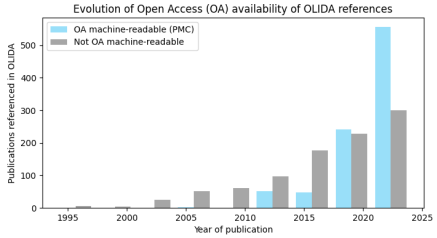
	Recall@1	Recall@3	Recall@5
Per entry	67%	72%	73%
Per reference	84%	88%	89%



Per OLIDA entry



Per OLIDA reference



Reasons for missed publications
Missing synonyms for non-SNP variants
Table parsing issues
Image recognition issues

Conclusion

Advantages of Variomes

Multi-variants and disease queries
Suitable for oligogenic diseases

Focus on recall
Normalization and expansion of the queries maximize the number of documents retrieved

Multiples collections
Ability to search in supplementary data and image-based contents in addition to abstracts and full texts

Areas for improvement

Limited access to publications
~Half of relevant publications are not Open Access

Non-SNP variants search
Missing synonyms for non-SNP variants

Parsing challenges
Difficulties with complex tables and character recognition from images